

Syed Aliyar Shah

Phone: (+92) 3359926750 (Cell) | Email address: syedaliyarshah@outlook.com | LinkedIn: [Profile](#) | Website: [Projects Portfolio](#)

WORK EXPERIENCE

30/08/2023 - CURRENT - IRVINE, CALIFORNIA, UNITED STATES

MECHANICAL DESIGN AND SIMULATION ENGINEER OHMITRON, INC

- **Projects: Electric Vehicles HV Li-ion Battery Packs — MegaWatt Charging Systems — Power Electronics Modules — Si-C based HV Inverters/Converters — 12V Li Batteries — On-Board Charger**
- Conducted advanced **CAD/CAE** based mechanical design, materials selection and multi-physics analysis to optimize structural, thermal and material behavior in SolidWorks, ANSYS, and ABAQUS for air, liquid and immersion cooled HV systems.
- Applied Finite Element Modeling and Analysis (**FEM&A**) and **continuum mechanics** approaches for structural stiffness, fatigue, impact, vibrational and design life analyses. Run multiple transient and steady-state thermal **CFDs** to evaluate thermo-mechanical coupling.
- Incorporated metal (Al, St), polymeric (ABS-PC), and GFRP **material systems** for enclosures, **structural** components and EM insulation. Evaluated materials manufacturability, cyclic mechanical and thermal durability by studying stress distributions and deformation plots.
- Performed design optimization techniques (**DFMEA, DVP&R, GD&T, DFA/DFM, DVT and PVT**) for **manufacturing** processes (injection moulding, 3d-printing, machining and sheet metal) and assessed material-process compatibility as per cost, performance and life.

24/06/2024 - 01/04/2025 - GUJRANWALA, PAKISTAN

RESEARCH AND DEVELOPMENT ENGINEER POLYMER PRIVATE INDUSTRIES

- **Thesis & Research Project: Design and Development of non-ceramic Glass fiber-epoxy Composite Insulator for Power Lines**
- Conducted indepth **composite mechanics R&D**, integrating material characterization, microstructural studies and design optimization for geometry under mechanical and electrical loads and established optimized pultrusion and hand-layup lines for manufacturing.
- Executed comprehensive **ASTM/ANSI** standardized experimentation including mechanical (tensile, flexural and creepage), electrical (dielectric and flashover), burn-out (fiber-resin volume and mass fractions) and surface hydrophobicity testings.
- Performed advanced structural **FEA** and composite material modeling in **ABAQUS** relating computational findings with experimental results to validate design assumptions. Investigated stress distributions, complex failure modes (**composite lamination and mixed failure theories**) and material behavior under real-world loadings.

01/06/2024 - 28/04/2025 - ISLAMABAD, PAKISTAN

RESEARCH ASSISTANT ADVANCED MANUFACTURING LAB | NUST

- **Conducted quantitative research on impact-resistance and failure mechanics of FDM-printed polymers** (PLA-WM, PLA, ABS), including repeated impact testing on BESMAK drop-weight apparatus and fracture-surface analysis.
- **Performed multi-material characterization and mechanics studies for polymer-matrix and fiber-reinforced (GFRP/CFRP) composites** for energy systems including burn-out and tensile/flexural testing for fiber/matrix fractions and mixed-mode failure analysis.
- **Developed and executed advanced FEA workflows in ANSYS and Abaqus**, incorporating composite layup modeling, cohesive interfaces, and progressive damage analysis. Built simulation networks to validate experimental data, enabling accurate prediction of impact behavior, delamination growth, and thermal-mechanical coupling in composite laminates and polymers across all research studies.
- **Initiated and co-authored multiple research studies**, including; *Assessment of Impact Resistance Characteristics of 3D-Printed Polymers* | *Comprehensive Characterizations of GFRP Composites for Overhead Insulator Applications* | *Multiphysics coupling simulation of mechanics-aging-dynamic loads of rubber materials* | *Design, Analysis, and Comparison of Hybrid & Non-Pneumatic Tires* | *Computational Approaches for Dynamic Crack Propagation* solidifying my interest in **composite mechanics and sustainable energy systems**.

25/04/2025 - 03/07/2025 - LAHORE, PAKISTAN

MECHANICAL DESIGN ENGINEER ENERTIA

- **Led design development of Heat Pump Water Heater** (Water Conditioners), **Pressure Vessels** and **Dispenser** for manufacturing.
- **Developed functional HVAC and thermal components** (capillary tubes, condenser, evaporator, valves, refrigerants) and increased efficiency by functionality testing & optimizing components through **thermal and CFD simulations in ANSYS**.

25/09/2025 - 01/11/2025 - BARCELONA, SPAIN

TRAINEE ENGINEER (CONTRACTUAL) PROCEEDIT

Learned manufacturing workflows and robot automation application using **Python** and **RPA** frameworks.

EDUCATION & TRAINING

21/11/2021 - 01/05/2025 - ISLAMABAD, PAKISTAN

BACHELORS IN MECHANICAL ENGINEERING- NATIONAL UNIVERSITY OF SCIENCE AND TECHNOLOGY (COLLEGE OF EME)

● SKILLS

Finite Element Analysis (FEA) | CAD/CAE/CAM | Laminate/RVE composites | Multi-physics modeling | Fluid Flow / CFD Simulations | Structural/Thermal Analysis | Research & Technical Writing | CATIA V5 / CATIA V6 | SolidWorks | ABAQUS | ANSYS | MATLAB | Origin Pro

Domains

Electric Vehicles and Energy Storage Systems | Structural Battery Concepts (BICS/MESC) | Composites and solids mechanics | Computational Mechanics | Nano-enhanced Materials

● PUBLICATIONS

Published Articles:

- [1] M. R. Khan, S. Ali, I. Asim, **S. A. Shah**, M. Ahmad, "Assessment of Impact Resistance Characteristics of 3D-Printed Polymers", *Key Engineering Materials*, 2025. DOI: [10.4028/p-g8owyW](https://doi.org/10.4028/p-g8owyW) .
- [2] S. Ali, **S. A. Shah**, M. Ahmad, S. M. Zain, A. R. Khan, "Design, Analysis and Comparison of Hybrid & Non-Pneumatic Tires", *ACI Avances En Ciencias E Ingenierías*, 2025. DOI: [10.18272/aci.3909](https://doi.org/10.18272/aci.3909) .

Submitted:

- [3] **S. A. Shah**, S. Ali, A. R. Khan, Z. Sajid, "Comprehensive Characterization of GFRP Composite Insulators for High-Voltage Overhead Lines", *Materials & Energy Systems Journal*. ([Manuscript](#))
- [4] Ali. S., **Shah. S. A.**, "High-Performance Composites for Aerospace, Hydrogen Fuel Storage, and Nuclear Technologies: A Review", *Advanced Materials Review*. ([Manuscript](#))

In Preparation:

- [5] **Shah. S. A.**, S. Ali, "Computational Approaches for Dynamic Crack Propagation: Review", *Mechanics of Materials*. ([Manuscript](#))
- [6] **Shah. S. A.**, M. R. Khan, "Experimental and Computational Validation of Mechanical and Failure Modes in GFRP Epoxy Composite Rods for HV Insulators", *Polymers*. ([Manuscript](#))
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● PROJECTS

- **High Voltage 85kW.hr EV Battery Pack** — design, FEA, CFD, prototyping and testing for structural, thermal and production optimization.
- **Design and Development of non-ceramic insulator for overhead lines (Thesis)**
- **FEM-based Fatigue Life Estimation of EV HV Battery Cell Holder Structures** — under cyclic mechanical and vibrational loading.
- **FEA of Shear Behavior in High-Stress Steel Bolted Beam Connections** — evaluated joint behavior and load transfer mechanisms.
- Visit [Mechanical Design Projects Portfolio](#) , and [Technical & Research Projects Document](#) , for other technical projects details.

● VOLUNTEERING

- **Bondh E Shams: [The Solar Water Project](#): Volunteer** in development and fund-raising drives for clean drinking water in rural areas.
- **Deputy Director at NUSTAG Automotive Group** - Conducted technical & soft-skills automotive training impacting over 500 students.
- **Deputy Director at ASME CEME Section** - Organized events and conducted workshops reaching over 1000 students on campus.

● HONOURS AND AWARDS

(1) Honors:

- **Gold Medalist (Best Project) for Final Year Project of Department of Mechanical Engineering 2025.**
- **1st Position at ASME E-Fests XRC Autonomous Vehicle International Competition 2023 and 2024.**
- **NUST High Achievers Award & Certificate of Merit (2023, 2024, 2025).**

(2) Awards:

- **1st Position at Mars Wheel and Wheel-Hub joint National Design Competition 2023 and 2024.**
- **3DxWorld24 by 3D Experience Dassault Systemes.**
- **Deputy Director at Formula SAE Project at NUSTAG Official and ASME CEME Student Section.**

(3) Certifications:

- **Certified SolidWorks Associate - Mechanical Design -Sustainability - Simulation - Additive Manufacturing** by Dassault Systemes.
- **Combinatorial Development of Amorphous Alloys** by UTP CCR, ISER.
- **Basics of Electric Vehicle Simulations using Ansys** by Skill Lync.
- For viewing all 25+ certificates, follow this [link](#) .

● RECOMMENDATIONS

(1) Mohamed Shaib (2) Dr. Zubair Sajid, (3) Dr. Muhammad Rehan Khan, (4) Dr. Syed Haris Iftikhar

- Recommendation details are attached in respective application section.